

Kristine Stanley

Senior Python Engineer | Backend, APIs & Cloud

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PROFESSIONAL SUMMARY

Senior Python Engineer with 11+ years delivering cybersecurity, connected-mobility, insurance, and learning-platform systems using **Python 3.5–3.12**, **FastAPI**, **Django**, **Flask**, **PostgreSQL**, **Redis**, **Kafka**, **AWS**, and **Azure**; specializes in secure API design, asynchronous processing, data-intensive services, legacy modernization, observability, and technical leadership that improves reliability, performance, release quality, and business scalability.

SKILLS

- **Languages:** Python 3.5–3.12, SQL, Bash, JavaScript, TypeScript, object-oriented programming, functional programming, asynchronous programming
- **Python Frameworks:** FastAPI, Django, Django REST Framework, Flask, SQLAlchemy, Pydantic, Alembic, Celery, asyncio, aiohttp, Pandas
- **API Technologies:** REST, GraphQL, OpenAPI, Swagger, WebSockets, Server-Sent Events, webhooks, OAuth 2.0, OpenID Connect, JWT, API versioning
- **Architecture:** Microservices, modular monoliths, event-driven architecture, domain-driven design, hexagonal architecture, CQRS, transactional outbox, saga workflows, multi-tenant systems
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB, DynamoDB, Redis, database modeling, schema migrations, indexing, query optimization, connection pooling
- **Messaging and Processing:** Apache Kafka, RabbitMQ, AWS SQS, Azure Service Bus, Celery, background workers, dead-letter queues, retries, event replay, idempotent consumers
- **Cloud Platforms:** AWS Lambda, ECS, EKS, API Gateway, S3, RDS, DynamoDB, SQS, SNS, CloudWatch; Azure Functions, App Service, AKS, Service Bus, Key Vault
- **Testing:** Pytest, unittest, Hypothesis, Factory Boy, Locust, Testcontainers, Postman, contract testing, integration testing, end-to-end testing
- **DevOps:** Docker, Kubernetes, Terraform, Helm, GitHub Actions, Azure DevOps, Jenkins, Poetry, pip, virtual environments, CI/CD pipelines
- **Monitoring:** OpenTelemetry, Datadog, Prometheus, Grafana, ELK Stack, Sentry, CloudWatch, structured logging, distributed tracing
- **Engineering Practices:** Technical leadership, architecture reviews, mentoring, code reviews, production support, incident response, root-cause analysis, Agile delivery

EXPERIENCE

Veracode Systems Inc — Senior Software Engineer

Remote/ Full time

April 2021 – May 2026

Somerville, MA

- Led backend architecture and delivery for enterprise application-security services using **Python 3.8–3.12**, **FastAPI**, **Django**, **PostgreSQL**, and **Kafka**, supporting vulnerability ingestion, policy evaluation, analytics, reporting, and remediation workflows.
- Replaced tightly coupled Django and worker modules with domain-oriented services, typed **Pydantic** contracts, dependency injection, and reusable infrastructure adapters, reducing production defects by **36%** across multiple engineering teams.
- Resolved severe API latency caused by repeated aggregate queries and oversized security payloads through composite indexes, cursor pagination, Redis caching, and query-plan analysis, improving P95 response time by **44%**.
- Modernized legacy Python services by removing unsupported dependencies, introducing strict typing with **mypy**, replacing blocking network calls with asynchronous clients, and adding automated compatibility checks for runtime upgrades.

- Designed secure multi-tenant APIs with **FastAPI**, OAuth 2.0, OpenID Connect, JWT validation, tenant-scoped repositories, and policy-based authorization to prevent unauthorized access to customer security data.
- Eliminated duplicate vulnerability findings produced during broker retries by implementing idempotency keys, transactional outbox processing, Kafka consumer checkpoints, and controlled dead-letter replay procedures.
- Built asynchronous ingestion pipelines using **Kafka**, **Celery**, AWS SQS, and bulk PostgreSQL operations, allowing burst traffic to be absorbed without overwhelming customer-facing services or downstream scanning systems.
- Introduced automated quality gates with **Pytest**, Hypothesis, contract tests, Testcontainers, and API integration suites, catching schema incompatibilities and migration errors before production deployment.
- Implemented distributed tracing and service-level monitoring with **OpenTelemetry**, **Datadog**, structured JSON logs, and actionable alerts, shortening diagnosis of memory leaks, queue backlogs, database contention, and external API failures.
- Mentored engineers through architecture reviews, production-readiness assessments, incident retrospectives, and performance-focused code reviews while coordinating technical decisions across product, security, frontend, and quality teams.

Cubic Telecom USA — Software Engineer

Remote/ Full time

January 2019 – April 2021

San Diego, CA

- Developed connected-mobility services using **Python 3.6–3.9**, **Django REST Framework**, **Flask**, **PostgreSQL**, and RabbitMQ, coordinating device registration, SIM provisioning, subscription changes, billing updates, and carrier synchronization.
- Improved provisioning throughput by **38%** through asynchronous task execution, bounded concurrency, batched database writes, and provider-specific rate controls that protected external telecom APIs during demand spikes.
- Resolved failed activations caused by inconsistent carrier identifiers and incompatible response schemas by introducing canonical domain models, typed adapters, correlation identifiers, and replayable reconciliation workflows.
- Designed REST APIs for device, account, billing, and connectivity operations with **OpenAPI**, request validation, versioned contracts, standardized error responses, and permission-aware endpoints for customer and operations applications.
- Implemented message-driven provisioning with **Celery**, RabbitMQ, durable queues, retry policies, dead-letter handling, and idempotent workers, preventing duplicate activations while preserving complete transaction traceability.
- Optimized slow device and usage queries through covering indexes, query prefetching, partition-aware access, connection-pool tuning, and removal of N+1 patterns as connected-device volumes increased.
- Containerized Python services with **Docker** and standardized CI/CD through **Azure DevOps**, adding unit tests, security scans, migration validation, and controlled promotion across development, staging, and production.
- Added structured logging, health endpoints, service metrics, and alert thresholds using **Azure Application Insights** and Grafana, helping support teams distinguish provider outages from internal application failures.
- Collaborated with telecom specialists, product managers, frontend engineers, and support teams to convert roaming and provisioning rules into maintainable service boundaries, documented APIs, and incremental delivery plans.

Asurion — Software Developer

Remote/ Full time

January 2017 – December 2018

Nashville, TN

- Built backend services for device-protection workflows using **Python 3.5–3.7**, **Django REST Framework**, **Flask**, Redis, and SQL Server, supporting policy eligibility, claims, payments, appointments, and repair-status tracking.
- Increased successful online claim completion by **27%** after redesigning orchestration APIs to preserve recoverable workflow state, return actionable validation details, and reduce unnecessary calls to slow downstream systems.
- Created a backend-for-frontend layer that combined customer, policy, inventory, payment, and appointment information, shielding web applications from incompatible contracts and partial service outages.
- Corrected duplicate repair appointments caused by repeated submissions and delayed responses through idempotency tokens, database uniqueness constraints, transactional validation, and clear request-state handling.
- Introduced **Redis** caching for frequently requested eligibility and repair-location data, applying explicit invalidation and short expiration rules to improve performance without returning outdated policy decisions.
- Developed **Celery** reconciliation jobs for incomplete claims and delayed payment callbacks, recovering records after provider timeouts while preserving audit history for customer-service and finance investigations.
- Expanded automated coverage with **Pytest**, unittest, Factory Boy, API integration tests, and service mocks for claims, payments, appointments, eligibility decisions, and recoverable failure scenarios.
- Supported production incidents by tracing requests across dependent systems, correcting malformed payloads and timeout behavior, and documenting root causes that prevented repeated customer-facing disruptions.

Rustici Software — Junior Software Developer

Hybrid/ Full time

June 2015 – December 2016

Nashville, TN

- Maintained learning-technology services using **Python 2.7 and 3.4**, **Django**, Flask, and MySQL, supporting course imports, SCORM launches, learner-state persistence, completion tracking, and xAPI statement processing.
- Implemented REST endpoints and background jobs for course ingestion and launch configuration, adding validation for malformed archives, incomplete manifests, and nonstandard learning-management-system payloads.
- Resolved memory growth during large course imports by streaming compressed files, limiting parallel extraction, releasing temporary resources, and recording actionable processing failures for support teams.
- Built defensive processing for duplicate and out-of-order learning events, preserving accurate completion, score, and resume state across platforms with inconsistent callback and retry behavior.
- Improved database reliability by adding targeted indexes, transactional updates, and repeatable migration scripts for learner-state tables used during high-volume course launches and reporting periods.
- Added automated tests around course imports, launch tokens, xAPI statements, and completion transitions, preventing regressions across multiple customer integrations and learning-platform implementations.
- Collaborated with senior engineers and support specialists to reproduce unusual SCORM and xAPI failures, document payload assumptions, and deliver fixes that shortened customer troubleshooting cycles.

EDUCATION

Tennessee State University — Bachelor's Degree in Computer Science

Sep 2011 – May 2015

CERTIFICATION

- **AWS Certified Developer – Associate** — Amazon Web Services
- **Microsoft Certified: Azure Developer Associate** — Microsoft
- **Certified Kubernetes Application Developer — CKAD** — The Linux Foundation and Cloud Native Computing Foundation